

Data acquisition

Figs inoculation (Control vs. 10^7 CFU/mL)

Capture of Hyperspectral Images (400-1000 nm)

Preprocessing

Read and reconstruction of scenes (.hdr, .raw)

Radiometric correction (Relative Reflectance)

Region of Interest Segmentation (ROI)

Feature Extraction

Extraction of mean (μ) and standard deviation (σ)

Optimization and Selection

Meta-heuristic Search (GA & PSO)

Fitness Evaluation (F1-Score with Light SVM)

Stop criteria
reached?

Yes

Fine tuning and Exhaustive Evaluation

Optimal Bands Subset Identification

NO

SHAP Analysis

Feature Importance Analysis

Summary Plot (Bee-swarm)